PORTFOLIO BUSINESS CASE

E-commerce Sales, Customer Behaviour and Profitability Analytics

Executive Business Report

Business Case	NovaCart Global
Analysis Period	2012-2015
Dataset	Global Superstore public sample
Delivery	Python, MySQL, Excel and Power BI

DISCLOSURE

NovaCart Global is a fictional company created for this portfolio project. The report uses the public Global Superstore sample dataset and should not be presented as verified transactional data from a named real company.

Executive Summary

DECISION FOCUS

The business is growing and profitable overall, but deep discounting and loss-making products reduce the quality of that growth. Discount governance is the clearest immediate action.

$12.64M	$1.47M	11.61%	25,739
Revenue	Profit	Profit Margin	Orders

796	178,312	$491.18	4.19%
Customers	Quantity	Average Order Value	Returned-Order Rate

What the analysis found

- Revenue grew from about $2.26M in 2012 to $4.30M in 2015, while profit grew from about $248.94K to $504.17K.
- Technology was the strongest category, while Tables produced a $64.08K loss and a -8.46% margin.
- Discount bands above 20% were loss-making overall; the above-50% band produced a $412.58K loss.
- Asia Pacific led revenue, while Europe produced the highest market profit and strongest margin.
- Repeat purchasers generated 96.86% of revenue, but this should not be described as formal retention.

Immediate management priorities

1. Control deep discounts with approval thresholds and expected-margin checks.
2. Review loss-making products, starting with Tables and the Bottom 5 products.
3. Assign country and regional profit reviews to the relevant Regional Managers.

1. Business Performance Overview

Revenue and profit increased across the four-year period. Growth was mainly volume-led: average order value decreased from about $500.32 in 2012 to $485.31 in 2015.

November and December produced the strongest revenue. However, October recorded the highest monthly margin at about 13.41%, while December margin was lower at about 10.73%.

MANAGEMENT INTERPRETATION

Peak sales periods should be evaluated using revenue, profit, margin and average order value together. High sales volume alone does not guarantee high-quality profit.



Figure 1. Power BI Executive Overview - overall KPIs, monthly trend, category performance and Market-to-Region analysis.

2. Product and Discount Risk

Issue	Validated evidence	Business meaning
Tables	$757.04K revenue; -$64.08K profit; -8.46% margin	High sales do not cover the cost and discount structure.
Loss-making products	677 products; -$247.84K combined profit	Revenue ranking alone can hide products that destroy value.
Discounts above 20%	Loss-making in aggregate	Discount approval should include an expected-margin check.
Discounts above 50%	$371.62K revenue; -$412.58K profit; -111.02% margin	The business lost more than the revenue generated in this band.

Statistical testing supported the discount finding. Discount and line-level margin had a Spearman correlation of -0.67, while discount band and loss status had a strong association (Cramer's V: 0.84).

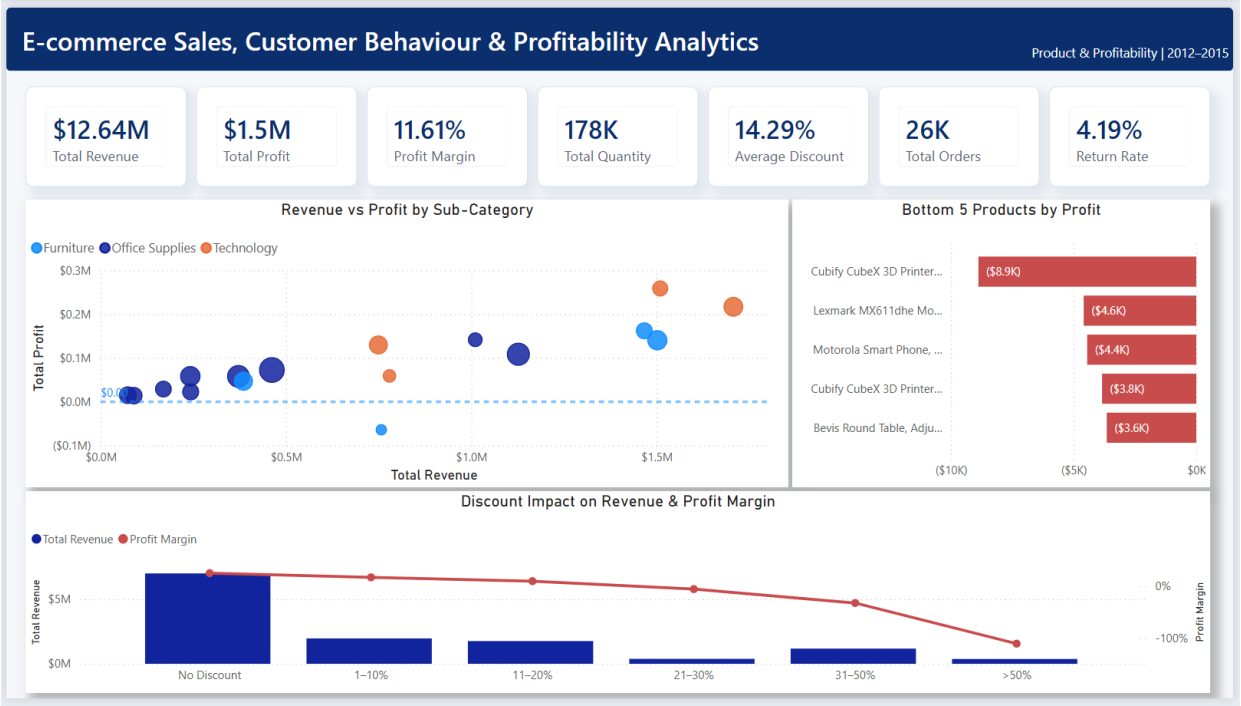


Figure 2. Product and Profitability page - sub-category performance, Bottom 5 products and discount impact.

3. Markets, Regions and Customers

Market performance

Market	Revenue	Profit	Margin	Avg. Discount
Asia Pacific	$4.04M	$403.18K	9.97%	18.10%
Europe	$3.29M	$449.55K	13.68%	9.09%
USCA	$2.36M	$304.22K	12.87%	15.04%
LATAM	$2.16M	$221.64K	10.24%	13.55%
Africa	$783.77K	$88.87K	11.34%	15.67%

Asia Pacific generated the most revenue, but Europe delivered the highest profit and strongest margin. Thirty-one countries were loss-making, so Market totals should be reviewed through Region and Country detail.

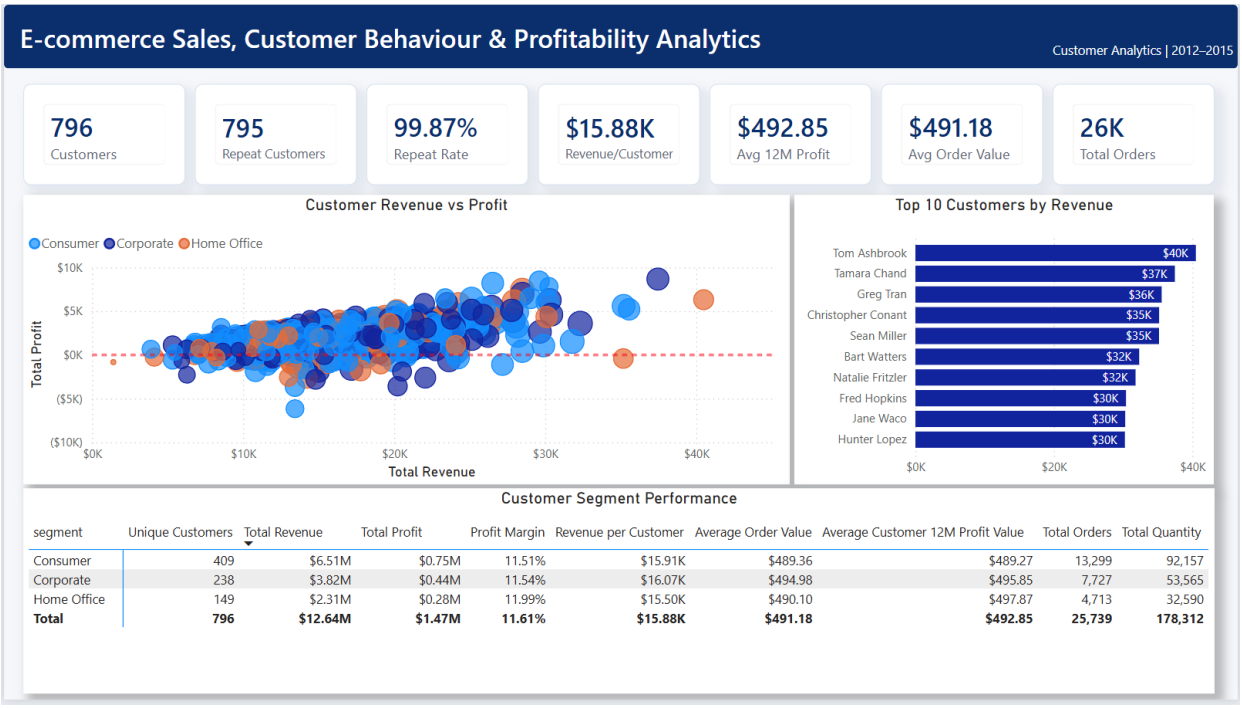
Customer segments

Segment	Customers	Revenue	Profit	Margin	Revenue / Customer
Consumer	409	$6.51M	$749.24K	11.51%	$15.91K
Corporate	238	$3.82M	$441.21K	11.54%	$16.07K
Home Office	149	$2.31M	$277.01K	11.99%	$15.50K

CUSTOMER FOCUS

The At Risk RFM group represented about $2.02M in historical revenue. Reactivation should focus on past value and expected profit, not blanket discounts.

4. Customer Behaviour, Returns and Advanced Analysis



segment	Unique Customers	Total Revenue	Total Profit	Profit Margin	Revenue per Customer	Average Order Value	Average Customer 12M Profit Value	Total Orders	Total Quantity
Consumer	409	$6.51M	$0.75M	11.51%	$15.91K	$489.36	$489.27	13,299	92,157
Corporate	238	$3.82M	$0.44M	11.54%	$16.07K	$494.98	$495.85	7,727	53,565
Home Office	149	$2.31M	$0.28M	11.99%	$15.50K	$490.10	$497.87	4,713	32,590
Total	**796**	**$12.64M**	**$1.47M**	**11.61%**	**$15.88K**	**$491.18**	**$492.85**	**25,739**	**178,312**

Figure 3. Customer Analytics page - customer profitability, Top 10 customers and segment performance.

Returns

The dataset contains 1,079 returned orders, equal to a 4.19% returned-order rate. USCA recorded the highest market rate at 4.47%, but market differences were small.

The above-50% discount band had a 3.73% returned-order rate, below the overall rate. This suggests returns were not the main reason for the deep-discount losses.

> **DATA LIMITATION**
> Returns are recorded at order level. Product-level findings must be described as products appearing in returned orders, not confirmed returned products.

Advanced analysis

- Furniture's product-level profitability was weaker than Office Supplies, but the discount effect was much larger than the category effect.
- Month-2 weighted customer activity was 28.19%; Month-12 activity was 38.35%. This is cohort activity, not subscription retention.
- Exact product pairs were too sparse for strong conclusions. Accessories and Phones showed a positive category-level signal across 377 orders.

5. Recommended Actions

Priority	Action	Suggested owner	Primary measure
1	Introduce discount approval bands and expected-margin checks.	Sales, Pricing, Finance	Margin and loss by discount band
2	Classify loss-making products as Keep, Fix or Exit; start with Tables.	Category, Procurement, Supply Chain	Negative-product loss; Tables margin
3	Review loss-making countries through Market, Region and Country drill-down.	Regional Managers, Pricing	Loss-making countries; regional profit
4	Reactivate high-value At Risk customers without blanket deep discounts.	CRM, Sales, Marketing	Reactivated customer profit
5	Capture exact returned product, return reason and return cost.	Operations, Customer Service	Return rate, reason and cost

Suggested 90-day plan

Timeframe	Main actions
Days 1-30	Approve discount rules, prepare the loss-making product list and build the At Risk customer list.
Days 31-60	Review Tables, test pricing and cost changes, analyse loss-making countries and improve return-data capture.
Days 61-90	Evaluate results, expand successful customer actions, test selected cross-selling ideas and assign regional follow-ups.

FIRST DECISION

Begin with discount governance. It is supported by the strongest statistical evidence and the largest directly observed loss pattern.

6. Decision Support Dashboard

The detailed Power BI page allows users to filter by Year, Market, Category and Region, review product-level profitability and connect the findings to the recommended actions.

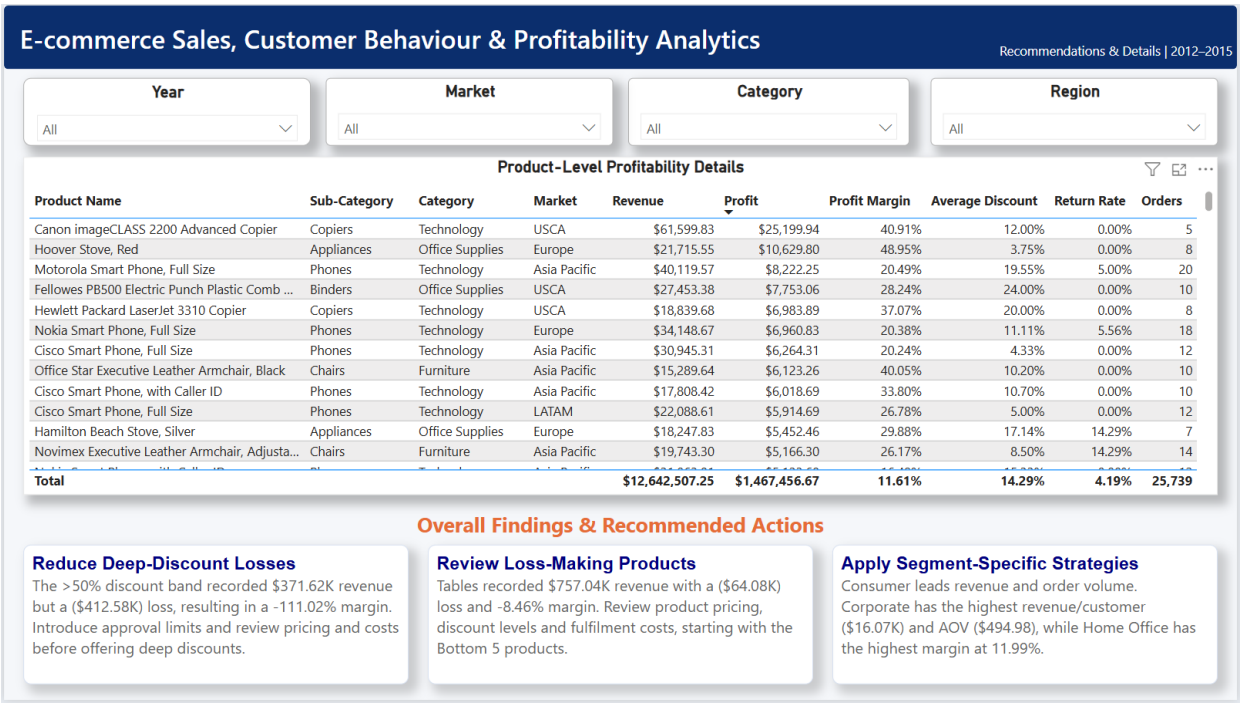


Figure 4. Recommendations and Details page - interactive product analysis and overall recommended actions.

Recommended review rhythm

- Weekly: monitor deep-discount exceptions and newly loss-making products.
- Monthly: review category, country, Region and customer profitability.
- Quarterly: reassess product Keep/Fix/Exit decisions and the impact of pricing changes.

7. Methodology and Analytical Model

The project follows an end-to-end analytics workflow. Raw data was preserved, cleaned outputs were stored separately and each analytical layer was validated before dashboard development.

Stage	Output
Python	Loading, quality audit, cleaning, EDA, statistical tests, cohort and basket analysis
MySQL 8	Star schema and 20 business-focused SQL queries
Excel	Power Query, Data Model, PivotTables, PivotCharts, slicers and KPI formulas
Power BI	Star schema, 19 organised DAX measures and four visible dashboard pages

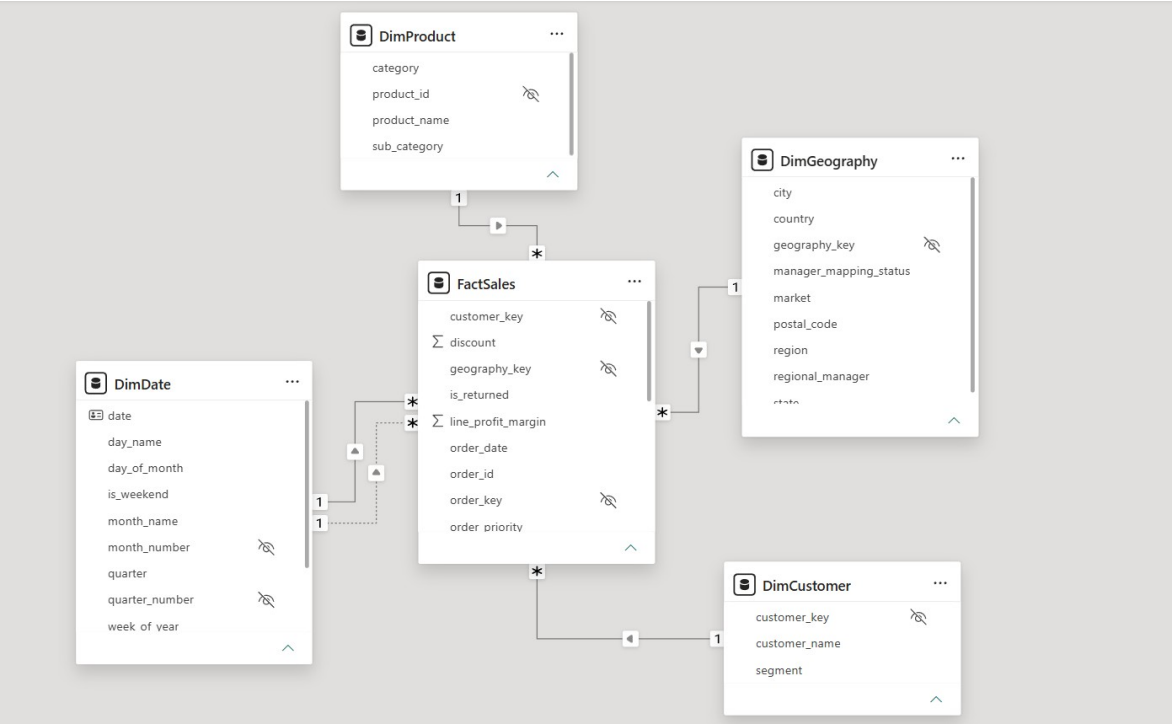


Figure 5. Power BI star schema with FactSales, DimDate, DimCustomer, DimProduct and DimGeography.

The cleaned Orders dataset contains 51,290 sales lines and 38 fields. The MySQL model contains 51,290 fact rows, 1,468 dates, 796 customers, 3,788 products and 3,856 geography records.

8. Limitations and Conclusion

Important limitations

- The project uses a public sample dataset and a fictional company scenario.
- Historical relationships do not prove direct cause and effect.
- Return information is available at order level, not individual returned-product level.
- The observed 99.87% repeat-purchase rate is not formal customer retention.
- The 12-month customer profit value is a historical proxy, not predictive lifetime value.
- Exact product-pair results were too sparse for strong basket conclusions.
- Missing postal codes were retained because they mainly belonged to non-US locations.
- Valid outliers and negative-profit transactions were retained because they are important to the business analysis.

Conclusion

NovaCart Global showed strong revenue and profit growth, but a meaningful share of sales came from low-margin or loss-making activity.

Deep discounting was the clearest risk, followed by loss-making products, Tables and unprofitable countries. Repeat customers remain important, and valuable At Risk customers provide a practical reactivation opportunity.

CLOSING MESSAGE

Future growth should be judged by profit and margin together with revenue. The existing Excel and Power BI dashboards provide the controls needed to monitor the action plan.

END OF REPORT